

# Holographic Memory for Zero-Shot Compositional Reasoning in Knowledge Graphs: A Mechanistic Study of Where and Why It Fails

Randhir Kumar<sup>1</sup>

<sup>1</sup>Independent Researcher , randhir2709vns@gmail.com

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## Abstract

Knowledge graph embedding (KGE) models such as TransE, DistMult, ComplEx and RotatE achieve strong single-hop link prediction but are not designed to answer *zero-shot compositional* queries: multi-hop questions whose relation chain was never observed during training. Holographic Reduced Representations (HRR), which bind and unbind symbols through circular convolution, are a theoretically appealing candidate for such reasoning because binding is approximately invertible and composable. We conduct a controlled study of two holographic memory variants—real-valued HRR and phase-only Fourier HRR (FHRR)—each paired with a modern Hopfield cleanup, on the full FB15k-237 test set over five random seeds. We report three findings. First, both variants are competitive *atomic* retrievers in the ranking sense: real HRR attains a filtered MRR of  $0.358 \pm 0.002$  and FHRR  $0.350 \pm 0.021$ , in the range of standard baselines (we stress this is not a parameter-matched comparison). Second, despite this, *both* variants fail at two-hop zero-shot composition: a one-sided binomial test cannot reject the hypothesis that their accuracy equals chance ( $p = 0.22$  and  $p = 0.85$  respectively), and accuracy is flat across all cleanup temperatures. Third, we characterise the FHRR failure mechanism and find it is not the magnitude (“modulus collapse”) story sometimes assumed for phasor memories: the cleanup re-projects onto the unit torus, so  $|z| = 1$  by construction. Two independent probes instead show the cleaned representation carries no phase information about the target—phasor cosine of  $-0.009 \pm 0.006$  to the true entity versus  $0.005 \pm 0.003$  to a random one, and a mean absolute phase error of 1.59 rad at the final hop, indistinguishable from the uniform value  $\pi/2$ . We resolve an apparent paradox—atomic ranking survives this near-random phase because it integrates aggregate codebook similarity, whereas the second bind is a per-component phase operation the cleanup does not support. An untrained control confirms training buys atomic competence but never composition. The limitation is architectural, and the cure must preserve phase coherence, not merely magnitude, across hops.

**Keywords:** Knowledge Graph Embeddings, Holographic Reduced Representations, Vector Symbolic Architectures, Compositional Reasoning, Modern Hopfield Networks, Zero-Shot Generalisation.

## 1 Introduction

Knowledge graphs (KGs) store facts as triples  $(h, r, t)$ , e.g.  $(\text{Marie\_Curie}, \text{nationality}, \text{Poland})$ . Knowledge graph embedding (KGE) methods learn vector representations of entities and relations so that plausible triples receive high scores [1, 2, 3, 4]. They are trained on, and evaluated against, *single-hop* queries: “given  $h$  and  $r$ , predict  $t$ ”.

Many questions of interest are inherently *compositional*: they chain several relations never jointly observed. “Which administrative regions lie in the country a given person is a national

of?” requires composing *nationality* with *contains*. If that composite never appears in training, a standard KGE model has no mechanism to infer it. We call this setting *zero-shot compositional reasoning*.

**Why holographic memory?** Holographic Reduced Representations (HRR) [5] and the broader family of Vector Symbolic Architectures (VSA) [6, 7] define an algebra over distributed vectors in which symbols are *bound* together and later *unbound*. The binding operator—circular convolution—is approximately invertible and associative, suggesting a memory storing facts as bound bundles could answer multi-hop queries by composing unbinding operations. Pairing this algebra with a modern Hopfield network [8] as an associative “cleanup” yields a natural candidate for zero-shot composition. Our central question is empirical: *does this candidate work, and if not, why?*

**Contributions.** All claims below are verified over five random seeds on the full FB15k-237 test set under a leakage-controlled protocol.

1. **Both holographic variants are competitive atomic retrievers** (in the ranking sense). Real HRR reaches filtered MRR  $0.358 \pm 0.002$  and FHRR  $0.350 \pm 0.021$ , in the range of standard single-hop baselines (Section 6.1). An ablation shows the Hopfield cleanup accounts for about half of the achieved accuracy. These are not parameter-matched comparisons.
2. **Both variants fail at zero-shot composition.** A one-sided binomial test against chance cannot be rejected for either model ( $p = 0.22$ ,  $p = 0.85$ ), and zero-shot accuracy is flat across all cleanup temperatures (Section 6.2). This is a negative result, reported plainly.
3. **We characterise the FHRR failure mechanism.** The magnitude-based “modulus collapse” intuition does not apply once the cleanup re-projects onto the unit torus ( $|z| = 1$ ). Two phase probes show the cleaned representation carries no directional information about the target, and we resolve the resulting paradox—atomic ranking survives near-random phase because it integrates aggregate similarity, whereas the second bind needs per-component phase coherence the cleanup does not supply (Section 6.3).

## 2 Related Work

**Knowledge graph embeddings.** TransE [1] models a relation as a translation  $h + r \approx t$ . DistMult [2] uses a bilinear diagonal score; ComplEx [3] extends it to complex embeddings for asymmetry; RotatE [4] represents relations as rotations,  $t \approx h \circ r$ . All are strong single-hop predictors with no native mechanism for composing unseen relation chains.

**Compositional and path-based reasoning.** The path-query framework of Guu et al. [9] composes relation embeddings along a path in vector space; NeuralLP [10] and RNNLogic [11] learn logical rules. These typically require supervision on paths or rules. We study the stricter zero-shot setting in which the composed relation is withheld entirely, and we do not claim to outperform trained compositional methods; we ask whether holographic binding *alone* composes.

**Holographic and vector-symbolic memory.** HRR [5] introduced circular convolution as a binding operator over real vectors; hyperdimensional computing [6] and VSA surveys [7] study related algebras, including the Fourier HRR (FHRR) variant on unit-modulus phasors with element-wise complex multiplication. The VSA literature discusses noise accumulation under repeated binding in general terms; here we measure, for an FHRR-with-Hopfield-cleanup pipeline specifically, *which* quantity (magnitude vs. phase) is lost.

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**Algorithm 1** Leakage-controlled zero-shot pair construction

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1: Input: train graph  $G_{\text{tr}}$ , test graph  $G_{\text{te}}$ , number of chains  $n$ 
2: For each ordered  $(r_1, r_2)$ , count chain train support: distinct mid-entities linking  $r_1$  then  $r_2$ 
   in  $G_{\text{tr}}$ 
3: In  $G_{\text{te}}$ , collect head–tail pairs reachable by each  $(r_1, r_2)$ 
4: Keep the  $n$  chains with most test pairs subject to chain train support  $\geq 100$ 
5: for each kept chain  $(r_1, r_2)$  and test pair  $(h, t)$  do
6:   if  $t$  is a direct train neighbour of  $h$  under any relation then
7:     discard  $(h, t)$   $\triangleright$  leakage
8:   else
9:     add  $(h, r_1, r_2, t)$  to the zero-shot set
10:  end if
11: end for
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**Modern Hopfield networks.** Ramsauer et al. [8] showed the update rule of continuous modern Hopfield networks coincides with attention and retrieves stored patterns via a softmax over similarities. We use this as the cleanup that maps a noisy unbound estimate back to a codebook entry.

### 3 Problem Formulation

Let  $E$  be a set of entities and  $R$  a set of relations; a knowledge graph is a triple set  $T \subseteq E \times R \times E$ , split into disjoint  $T_{\text{tr}}, T_{\text{va}}, T_{\text{te}}$ .

**Single-hop (atomic) task.** Given  $(h, r)$ , predict  $t$  with  $(h, r, t) \in T_{\text{te}}$ . We rank all candidate tails and report filtered metrics.

**Two-hop compositional task.** For a chain  $(r_1, r_2)$ , the compositional answer set for head  $h$  is

$$A_{h,r_1,r_2} = \{t \in E : \exists m \in E, (h, r_1, m) \in T \wedge (m, r_2, t) \in T\}. \quad (1)$$

The model predicts an element of  $A_{h,r_1,r_2}$  without having been trained on the composite  $r_1 \circ r_2$  and without an explicit representation of the intermediate entity  $m$ .

**Zero-shot protocol and leakage control.** We extract the ten most frequent two-hop chains in FB15k-237 (Algorithm 1). “Train support” for a chain  $(r_1, r_2)$  is the number of distinct mid-entities  $m$  such that some  $(h, r_1, m)$  and  $(m, r_2, t)$  both occur in training; we require it to be  $\geq 100$ . The individual relations  $r_1, r_2$  are themselves seen atomically in training; only their *composition* is withheld. Chains with a repeated relation ( $r_1 = r_2$ ) are permitted and handled identically, since the intermediate-entity constraint in  $A_{h,r_1,r_2}$  is defined per hop and does not assume  $r_1 \neq r_2$ . To avoid trivial answers we remove every  $(h, t)$  for which  $t$  is directly reachable from  $h$  via any single training relation. This removes 98 pairs (0.14%), leaving 69,855 genuinely zero-shot pairs. Chance accuracy is  $1/|E| \approx 6.8 \times 10^{-5}$ .

## 4 Method

### 4.1 Holographic algebra

For  $a, b \in \mathbb{R}^D$ , circular convolution binds them,

$$(a \circledast b)_k = \sum_{j=0}^{D-1} a_j b_{(k-j) \bmod D}, \quad \widehat{a \circledast b} = \hat{a} \odot \hat{b}, \quad (2)$$

with  $\hat{\cdot}$  the DFT and  $\odot$  element-wise product ( $O(D \log D)$  via the FFT). Unbinding uses the conjugate,  $a \approx (a \circledast b) \oslash b = \mathcal{F}^{-1}(\widehat{a \circledast b} \odot \bar{\hat{b}})$ . Facts are superposed into one memory vector,  $M = \sum_{(h,r,t) \in T_{\text{tr}}} e_h \circledast \rho_r \circledast e_t$ , with entity embeddings  $e_x$  and role embeddings  $\rho_r$ .

### 4.2 Hopfield cleanup

Unbinding from a superposed memory is noisy. A modern Hopfield cleanup maps the estimate back to the entity codebook  $E \in \mathbb{R}^{|E| \times D}$ ,

$$\text{clean}(z) = \text{softmax}(\beta \langle z, E \rangle) E, \quad (3)$$

with inverse temperature  $\beta$ ; a hard variant uses straight-through  $\arg \max$ .

### 4.3 Real HRR and Complex FHRR

**Real HRR** uses real embeddings  $e_x, \rho_r \in \mathbb{R}^D$ ,  $D=1024$ . Atomic prediction is  $\hat{t} = \text{clean}(M \oslash (e_h \circledast \rho_r))$  scored against  $E$ ; the two-hop prediction unbinds twice, cleaning the intermediate (hard) before the second bind. **Complex FHRR** represents each symbol by phases  $\phi \in [-\pi, \pi]^D$ ,  $D=512$ , as a unit phasor  $z = e^{i\phi}$ . Binding is element-wise complex multiplication (phase addition); unbinding multiplies by the conjugate. Because FHRR embeddings are complex,  $D=512$  complex coordinates equal 1024 real numbers—the same real parameter budget as Real HRR, so the two models are roughly parameter-matched to each other. Crucially, the FHRR cleanup extracts the phase of the convex combination and re-embeds it as  $e^{i\phi}$ , *forcing*  $|z| = 1$  at every cleanup output—central to Section 6.3. Figure 1 shows both flows.

### 4.4 Training objective

Both models minimise a cross-entropy atomic loss plus a contrastive invertibility regulariser  $\mathcal{L} = \mathcal{L}_{\text{atom}} + \lambda \mathcal{L}_{\text{inv}}$ ,  $\lambda=0.2$ , where  $\mathcal{L}_{\text{inv}}$  encourages  $f \circledast r$  to unbind back to  $f$  and  $r$ . The memory  $M$  is Hebbian initialised and refined by gradient descent. We use Adam (lr  $10^{-3}$ , cosine-annealed to  $10^{-5}$ ), batch 2048, gradient clip 1.0.

## 5 Experimental Setup

**Dataset.** Canonical FB15k-237: 272,115 train, 17,535 valid, 20,466 test triples;  $|E| = 14,778$  entities (entities and relations share one index space) and 237 relations.

**Metrics.** *Top-1 accuracy* is the fraction of test queries whose top-scored candidate is the gold tail. We also report filtered *MRR* and *Hits@{1, 3, 10}*, removing all other true tails before scoring the gold tail. Crucially, all ranking metrics are computed on the *full* test set (20,466 queries), not a sample. For composition we report top-1 zero-shot accuracy over the pairs of Algorithm 1 and a one-sided binomial test against chance.

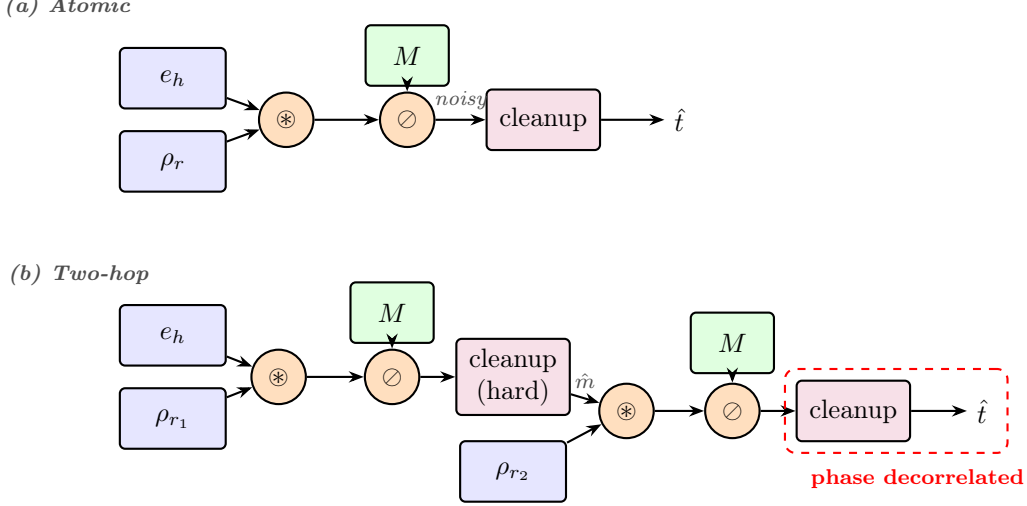


Figure 1: Holographic memory architecture. **(a)** An atomic query binds head  $e_h$  with role  $\rho_r$ , unbinds from the superposed memory  $M$ , and projects to an entity via the Hopfield cleanup. **(b)** A two-hop query repeats bind–unbind–cleanup: the first (hard) cleanup yields an estimated intermediate  $\hat{m}$ , which is bound with the second role and unbound again. The dashed red box marks the failure point—after the second hop the cleaned representation’s phase is decorrelated from the true target (Section 6.3).

**Reproducibility.** All headline numbers are mean  $\pm$  std over seeds  $\{1, 2, 3, 4, 42\}$ . Real HRR uses  $D=1024$ ,  $\beta=8$ ; FHRR  $D=512$  complex ( $= 1024$  real),  $\beta=12$ .

## 6 Results

### 6.1 Both variants are competitive atomic retrievers

Table 1 reports full-test-set single-hop performance over five seeds, visualised in Figure 2. Real HRR reaches filtered MRR  $0.358 \pm 0.002$  and FHRR  $0.350 \pm 0.021$ , both in the range of standard baselines, with Hits@10 around 0.53. We emphasise the baseline figures are representative literature values at their own dimensionalities and evaluation conventions; they are *not* a parameter-matched re-implementation, and our models additionally carry a Hebbian memory  $M$ . Table 1 should be read as evidence that holographic retrieval *can be competitive*, not as a controlled ranking, and we do not claim to beat any baseline. Notably FHRR—whose top-1 accuracy (0.126) is below real HRR’s (0.158)—attains a comparable *MRR* at the same real-parameter budget, indicating its errors are near-misses in the ranking rather than gross failures.

**The Hopfield cleanup carries half the performance.** Replacing the cleanup of Eq. (3) with a direct normalisation of the unbound estimate (*no Hopfield*) drops real-HRR atomic accuracy from 0.158 to 0.081—a relative decrease of  $\approx 48\%$  (Table 2). The cleanup is essential, not cosmetic. We do not ablate the FHRR cleanup separately because its phase re-projection is architecturally mandatory: without it the representation leaves the unit torus and binding is no longer well defined.

### 6.2 Both variants fail at zero-shot composition

In sharp contrast to atomic performance, two-hop zero-shot accuracy is at or near chance for *both* variants. Mean accuracy is  $1.7 \times 10^{-4}$  (real HRR) and  $2.9 \times 10^{-5}$  (FHRR), against chance  $6.8 \times 10^{-5}$ . We run a one-sided binomial test per seed ( $H_0$ : accuracy = chance). For FHRR

Table 1: Atomic (single-hop) link prediction on FB15k-237, filtered setting, *full test set* (20,466 queries). Our values are mean  $\pm$  std over five seeds. Baseline values are representative literature figures at their own dimensionalities, for context only—not a parameter-matched comparison (Section 7). Top-1 is reported for our models as a diagnostic that motivates the margin analysis in Section 6.3; it is typically not reported for the baselines and is left blank there.

| Model                     | Top-1             | MRR               | Hits@1            | Hits@3            | Hits@10           |
|---------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| TransE [1] <sup>†</sup>   | —                 | 0.294             | —                 | —                 | 0.465             |
| DistMult [2] <sup>†</sup> | —                 | 0.241             | —                 | —                 | 0.419             |
| ComplEx [3] <sup>†</sup>  | —                 | 0.247             | —                 | —                 | 0.428             |
| RotatE [4] <sup>†</sup>   | —                 | 0.338             | 0.241             | 0.375             | 0.533             |
| Real HRR (ours)           | 0.158 $\pm$ 0.001 | 0.358 $\pm$ 0.002 | 0.267 $\pm$ 0.002 | 0.392 $\pm$ 0.003 | 0.540 $\pm$ 0.003 |
| Complex FHRR (ours)       | 0.126 $\pm$ 0.001 | 0.350 $\pm$ 0.021 | 0.262 $\pm$ 0.017 | 0.390 $\pm$ 0.024 | 0.524 $\pm$ 0.028 |

<sup>†</sup> Representative literature values; not re-run here.

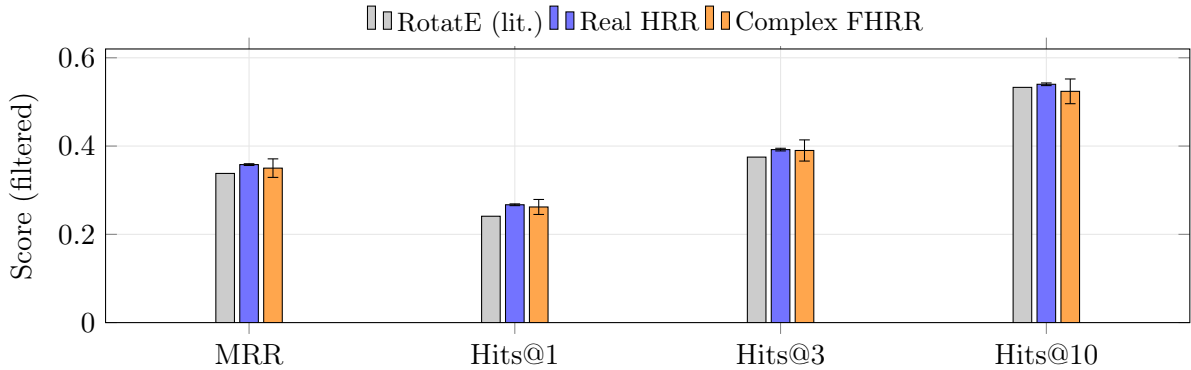


Figure 2: Atomic link-prediction metrics on the full FB15k-237 test set. Both holographic models are in the range of the RotatE reference (grey, literature value, not parameter-matched). Error bars are  $\pm$  std over five seeds; the RotatE bar is a single literature point estimate and has no error bar available.

the null is not rejected at  $\alpha = 0.05$  for any of the five seeds (per-seed  $p$  in  $[0.34, 1.0]$ ): there is no evidence FHRR exceeds chance. For real HRR the picture is marginal—the null is rejected for three of five seeds (per-seed  $p$  in  $[0.0000, 0.99]$ )—but the absolute accuracy remains tiny ( $\sim 1.7 \times 10^{-4}$ , i.e. a handful of the 69,855 pairs), so even where it is statistically above chance it is far from any usable signal and orders of magnitude below atomic performance. We therefore describe both models as failing the compositional task. The failure is not a temperature artifact—sweeping  $\beta \in \{1, 5, 10, 20, 50\}$  leaves accuracy flat and near chance (Figure 3). No temperature elicits composition, indicating a structural obstacle.

### 6.3 The FHRR failure mechanism: phase, not magnitude

A common intuition for phasor memories under iterated binding is *modulus collapse*: the softmax cleanup of Eq. (3) is a convex combination of unit phasors, so its output should have  $|z| \ll 1$ , decaying toward a noise floor over hops. We tested this directly and found it inapplicable here.

**The modulus probe is uninformative by construction.** The mean per-component modulus of the cleaned representation is  $|z| = 1.0000$  (std 0.0000, min = max = 1.0000) at *both* hops, across all five seeds. The reason is architectural: the FHRR cleanup extracts the phase of the convex combination and re-embeds it as a unit phasor  $e^{i\phi}$ , so  $|z| = 1$  *by construction*. Any

Table 2: Core-component ablation (Real HRR, five seeds, full test set). Removing the Hopfield cleanup nearly halves atomic accuracy; zero-shot accuracy is at chance in both cases.

| Configuration      | Atomic Top-1      | Zero-Shot             |
|--------------------|-------------------|-----------------------|
| Full model         | $0.158 \pm 0.001$ | $0.00017 \pm 0.00009$ |
| – Hopfield cleanup | $0.081 \pm 0.010$ | $0.00030 \pm 0.00010$ |

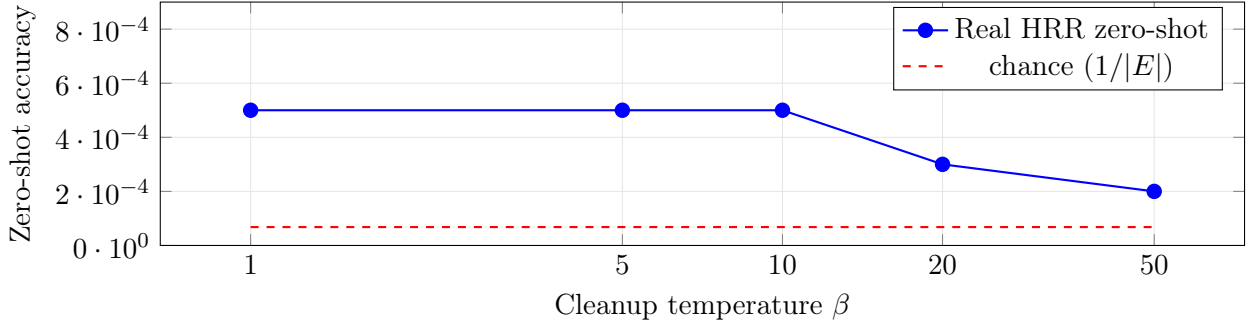


Figure 3: Cleanup temperature sweep (Real HRR, representative seed). Zero-shot accuracy stays near the chance line (red) for all  $\beta$ ; no temperature elicits composition.

magnitude-based diagnostic is therefore vacuous for this model; the modulus-collapse narrative does not apply once cleanup re-normalises onto the unit torus.

**Probe 1 — phase coherence.** Since magnitude is fixed, only the phase can carry information. We measure the phasor cosine similarity between the cleaned two-hop representation and (i) the *true* target and (ii) a *random* entity (Figure 4). Across five seeds, similarity to the true target is  $-0.009 \pm 0.006$ —statistically indistinguishable from, and on average slightly below, the  $0.005 \pm 0.003$  to a random entity. Both are effectively zero: the cleaned representation retains *no* directional information about the answer.

**Probe 2 — phase error propagation.** We measure the per-component absolute phase error between the cleaned representation and the true entity at each hop (Figure 5). The mean absolute error is 1.54rad at hop 1 and 1.59rad at hop 2, essentially equal to the uniform value  $\pi/2 \approx 1.571$ . The fraction of components with error beyond  $\pi/2$ —where cosine is uninformative—is 0.48 and 0.51, and the circular variance is 0.97 and 0.99 (uniform = 1). The cleaned phase is close to uniform-random already at the first hop and fully so by the second. (We note that the hop-1 measurement requires a ground-truth intermediate entity; when a head reaches several mid-entities via  $r_1$  we use the first one that also completes the chain to the gold target, so the hop-1 number carries mild ambiguity for high-cardinality chains. The hop-2 measurement against the gold target, which drives our conclusion, does not share this ambiguity.)

**Probes 3–4 — renormalisation and hard cleanup do not help.** Two interventions that would rescue a *magnitude*-driven failure have no effect on a *phase*-driven one. Explicitly re-normalising intermediate and final representations to unit modulus leaves zero-shot accuracy at 0.000000; replacing the soft cleanup with a hard straight-through arg max at both hops likewise yields 0.000000. Neither magnitude restoration nor sharper selection recovers the lost phase.

**The paradox: atomic ranking survives near-random phase.** A puzzle remains: how can FHRR achieve MRR 0.350 when even the first-hop cleaned phase is near uniform? The



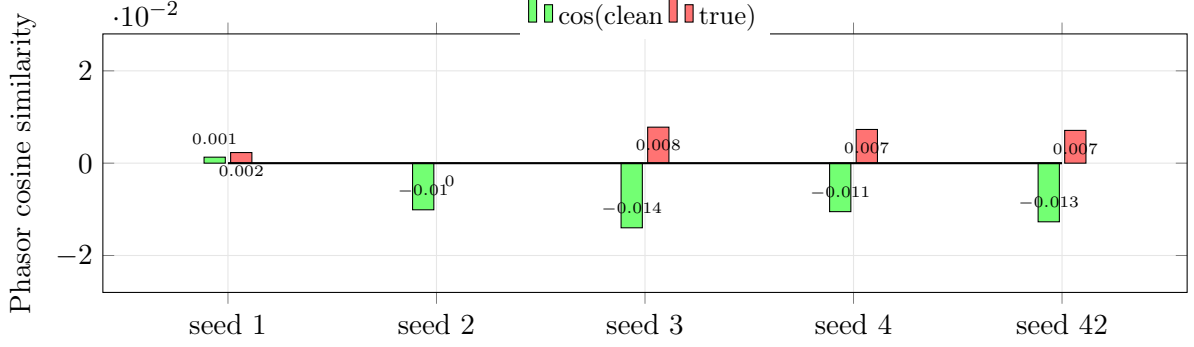


Figure 4: FHRR phase coherence per seed: phasor cosine of the cleaned two-hop representation to the true target (green) vs. a random target (red). Both hover around zero and are statistically indistinguishable—direct evidence of phase decorrelation.

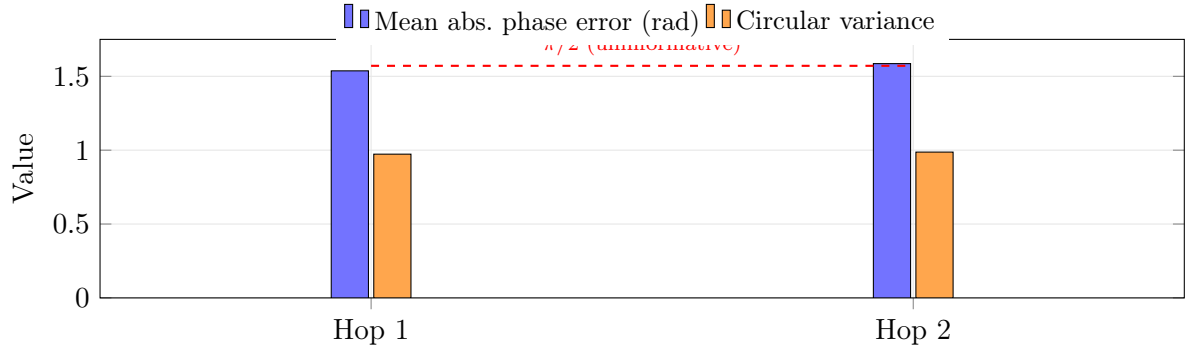


Figure 5: Phase error propagation. Mean absolute phase error (blue) sits at the uniform value  $\pi/2$  (red dashed) at both hops; circular variance (orange) approaches its uniform limit of 1. The cleaned phase carries essentially no target information at either hop.

resolution is that the two tasks use different signals. We measured the atomic softmax *margin* (gold logit minus the largest incorrect logit). It is negative on average ( $-1.82$  for FHRR,  $-1.92$  for real HRR), and the gold entity is the single top choice only  $\approx 13\text{--}16\%$  of the time—consistent with top-1 accuracy. Yet MRR is high (0.35), so the gold entity is reliably ranked among the top few even when not first. In other words, atomic competence is a *ranking* property recovered by integrating aggregate codebook similarity across all  $D$  dimensions, robust to per-component phase noise. Composition, by contrast, feeds the cleaned intermediate back into a *bind*, a per-component phase operation: it requires each component’s phase to be correct, not merely the aggregate similarity to be peaked. The cleanup supplies the latter but not the former, so single-hop ranking survives while the second bind operates on phase noise.

**An untrained control isolates training from architecture.** A freshly initialised FHRR has atomic accuracy  $\approx 0$  while the trained model reaches  $0.126 \pm 0.001$ —training clearly helps single-hop retrieval. Yet both are equally at chance on the two-hop task. Optimisation buys atomic competence but never composition; the limitation is a property of the architecture, not of the training run.

## 7 Discussion

**Why the phase is lost.** Each unbinding from the superposed memory  $M$  injects cross-talk from every other stored fact. The Hopfield cleanup is strong enough to rank the correct



entity from this noisy estimate—hence atomic retrieval works in the ranking sense—but the phasor it emits is reconstructed from a softmax-weighted average whose per-component phase is close to uniform. Binding is a per-component phase operation, so feeding this phase-incoherent intermediate into the second bind compounds noise rather than signal, and the second unbinding adds a fresh layer of cross-talk. By the final cleanup the angular signal that should point at the target has been averaged away. The unit-modulus constraint that makes FHRR elegant for a single binding is precisely what removes the magnitude headroom that might otherwise have flagged low confidence, leaving a confident-looking but directionless phasor.

**Real HRR fails too.** Real HRR also fails at two-hop composition. We did not isolate a single clean mechanism comparable to the FHRR phase analysis; the likeliest explanation is the same accumulation of memory cross-talk across hops, here as additive noise rather than phase mixing. We avoid over-attributing the small atomic gap between the models, since they also differ in representation; a full mechanistic account of real-HRR composition failure is left open.

**Implications.** Our findings caution against two intuitions. First, competitive single-hop performance of a holographic memory does *not* imply any capacity for zero-shot composition; the two are decoupled here. Second, the “modulus collapse” picture is not universal: once a cleanup re-normalises onto the unit torus, magnitude is fixed and the relevant failure is in the phase. Architectures intending to compose with phasor memories will need mechanisms that explicitly preserve phase coherence across hops—e.g. per-hop error correction, confidence-aware cleanup that retains magnitude, or memory structures that reduce inter-fact cross-talk.

**Limitations.** This study is deliberately controlled and therefore narrow. (i) We evaluate on a single real dataset (FB15k-237); behaviour on sparser graphs (WN18RR) or small ontologies (UMLS) may differ and is future work. (ii) The baseline figures in Table 1 are representative literature values, not a controlled re-implementation; the atomic comparison is indicative only, and we do not claim to beat any baseline. (iii) We do not compare against trained compositional methods (e.g. Guu et al. [9]); our question is whether holographic binding *alone* composes, and the answer is negative. (iv) We study two-hop chains over ten high-frequency relation pairs; deeper or more diverse compositions, and a breakdown by relation cardinality (1-1/1-N/N-N), are untested. (v) The hop-1 phase-error measurement uses an approximate ground-truth intermediate when a head reaches multiple mid-entities; the hop-2 measurement, which drives our conclusion, does not. None of these caveats affects the central result—chance-level composition with a phase-localised mechanism for FHRR—but they bound its scope.

## 8 Conclusion

We studied holographic memory with a modern Hopfield cleanup as a candidate for zero-shot compositional reasoning in knowledge graphs. Both real and Fourier variants proved competitive single-hop retrievers in the ranking sense (filtered MRR 0.358 and 0.350), with the Hopfield cleanup responsible for about half of the achieved accuracy. Yet both failed at two-hop zero-shot composition, at statistically chance accuracy and independent of cleanup temperature. Our main contribution is mechanistic: the commonly assumed modulus-collapse explanation is inapplicable to a cleanup that re-normalises onto the unit torus, and *phase decorrelation* is the operative failure mode for FHRR—the cleaned two-hop representation carries no more target information than a random entity, with phase error at the uniform  $\pi/2$  limit. We resolve the apparent paradox that atomic ranking survives this near-random phase: single-hop readout integrates aggregate similarity, whereas the second bind demands per-component phase coherence the cleanup does not provide. We hope this sharpens the target for future compositional reasoners: preserve phase coherence, not merely magnitude, across hops.

## Acknowledgments

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## Reproducibility Statement

All results derive from five fixed-seed checkpoints ( $\{1, 2, 3, 4, 42\}$ ) under the leakage-controlled protocol of Algorithm 1. Atomic metrics use the filtered convention on the full 20,466-query test set. The phase-coherence and phase-error probes, the modulus probe, the softmax-margin analysis, the temperature sweep, the binomial test, and the untrained control are all computed by inference over the stored checkpoints. Hyperparameters are in Section 5. Code and configurations will be released at <https://github.com/iamhero2709/holographic-memory>.

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